PART 1: DATA UNDERSTANDING

ShopSmart Retail Ltd | AnalystLab Africa Week 1, Mission 01 | Prepared by Godfrey Miracle

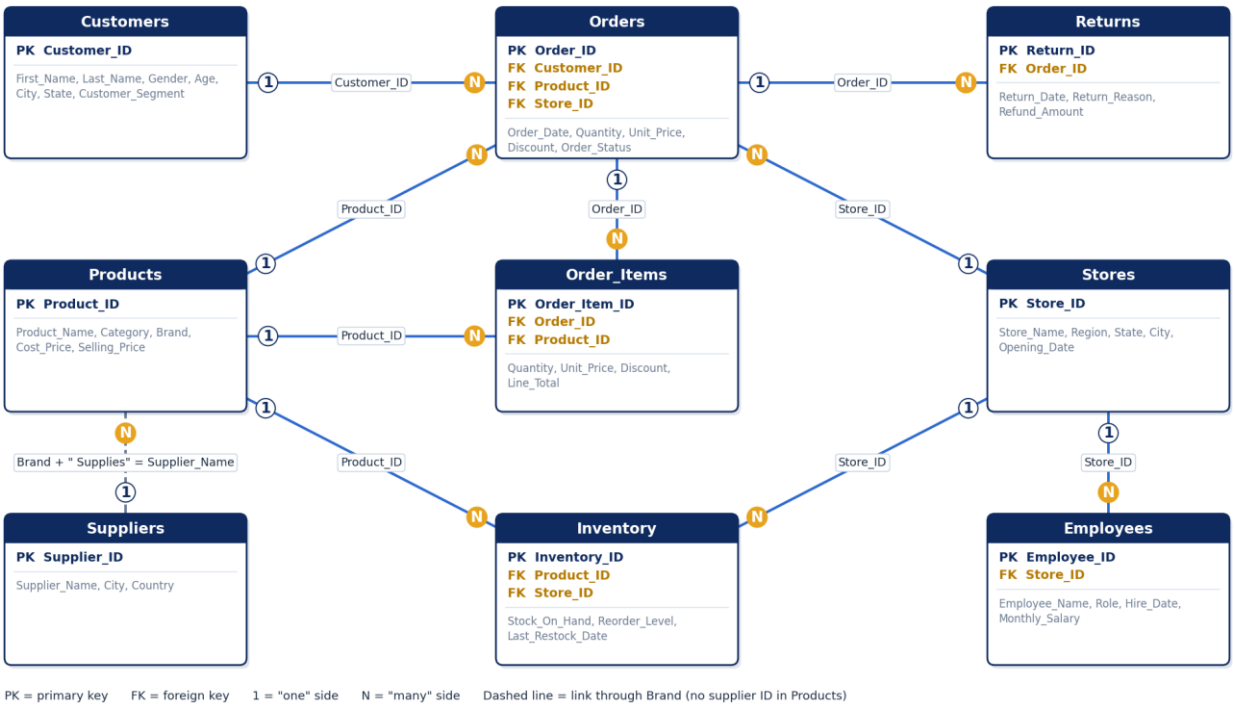
1. How many datasets were provided?

Nine data tables: Customers, Products, Orders, Order_Items, Stores, Returns, Employees, Suppliers and Inventory. The assignment also lists a Data Dictionary workbook, but it was not included in the workbook supplied, so column meanings were taken from the column names and values.

2 to 4. What each dataset represents, with its primary and foreign keys

Dataset	What it represents	Rows	Primary key	Foreign key(s)
Customers	People who buy from ShopSmart: name, gender, age, location, registration date and segment (Bronze, Silver, Gold)	5,000	**Customer_ID**	None
Products	The catalogue of items for sale: name, category, brand, cost price and selling price	500	**Product_ID**	None (Brand links to Suppliers)
Orders	One row per purchase: who bought which product, in which store and when, with quantity, price, discount and status	50,000	**Order_ID**	Customer_ID, Product_ID, Store_ID
Order_Items	Line detail for each order, including Line_Total (one line per order in this data)	50,000	**Order_Item_ID**	Order_ID, Product_ID
Stores	The shop locations, with region, state, city and opening date	25	**Store_ID**	None
Returns	Orders that customers sent back, with return date, reason and refund amount	2,500	**Return_ID**	Order_ID
Employees	Staff and the store where each one works, with role, hire date and salary	125	**Employee_ID**	Store_ID
Suppliers	The companies that supply the goods, with contact details and country	489	**Supplier_ID**	None (linked to Products through Brand)
Inventory	Stock held for each product in a store: stock on hand and reorder level	500	**Inventory_ID**	Product_ID, Store_ID

5. Relationship between the tables



- **How the tables connect:** each customer, product and store appears in many orders; each order has an order line and can have returns; each store has many employees and stock records; each product has stock records.
- **Suppliers:** Products has no supplier ID, so each product's Brand plus the word “Supplies” is matched to the supplier name (dashed line).
- **Checks:** every primary key is unique and every foreign key value exists in its parent table, so there are no orphan records. Order_Items repeats Orders one-to-one in this data.

This document answers the ten business questions in Part 8 of the assignment. All figures come from the pivot tables in ShopSmart_Analysis.xlsx. Amounts are in naira. The assumptions behind the numbers are listed at the end.

1. Which product generated the highest revenue?

> **Answer: Home Item 400 generated the highest revenue: ₦21,638,851.**

Evidence: Next are Groceries Item 115 (₦21,429,952) and Groceries Item 331 (₦20,809,181). The top 10 products together account for only about 2.7% of total revenue, so revenue is spread across the whole 500-product catalogue rather than driven by a few hits.

Source in workbook: Pivots: Top 10 Products

2. Which store performed best?

> **Answer: STR015 (ShopSmart Lagos) performed best with ₦311,890,852 in revenue.**

Evidence: Next are STR021 Lagos (₦311,156,234) and STR017 Abuja (₦309,777,616). The lowest store, STR023 Abuja, made ₦294,223,781, so the best store is only 6.0% ahead of the weakest. Performance is very even across the 25 stores. Stores were compared by Store_ID because the 25 stores share only 10 names.

Source in workbook: Pivots: Sales by Store; Dashboard: Revenue by Store

3. Which category sells the most?

> **Answer: Beauty sells the most, with ₦1,762,500,500 (23.3% of revenue) and also the most orders (11,809).**

Evidence: Groceries ₦1.58B (20.8%), Electronics ₦1.53B (20.2%), Fashion ₦1.41B (18.7%), Home ₦1.29B (17.1%). Beauty also has the most products (119), so its lead reflects range size: per product it earns less than Groceries or Electronics.

Source in workbook: Pivots: Sales by Category; Dashboard: Revenue by Category

4. Which month generated the highest revenue?

> **Answer: December 2025 generated the highest revenue: ₦660,629,527.**

Evidence: The lowest month was February 2025 (₦566,592,552). February had fewer orders (3,816 against roughly 4,200 in a typical month) and only 28 days. Per day it earned about ₦20.2M against ₦20.4M to ₦21.3M in other months, so the dip is mostly explained by the shorter month rather than weak trading.

Source in workbook: Pivots: Sales by Month; Dashboard: Monthly Sales Trend

5. Which customers spend the most?

> **Answer: Margaret Hawkins (CUST02989) spends the most: ₦4,343,570, followed by Andrew Martin (CUST00883, ₦4,230,195) and Joseph Collins (CUST00904, ₦4,088,527).**

Evidence: The top 10 customers together spent ₦39,719,559, only about 0.5% of total revenue, so there are no dominant big spenders. The top 10 include 5 Gold, 4 Silver and 1 Bronze member, so Gold members are slightly over-represented, but the sample is small.

Source in workbook: Pivots: Top 10 Customers

6. Which products are frequently returned?

> **Answer: Home Item 263 (11 returned orders, 11.6% return rate), Beauty Item 262 (10, 11.4%), Beauty Item 443 (10, 10.9%), Fashion Item 349 (10, 10.6%) and Home Item 5 (10, 10.4%) are returned most often.**

Evidence: The overall return rate is 4.88%. Each product has only about 100 orders and an average of about 5 returned orders, so these counts are small. The products should be reviewed, but the data does not prove a quality problem. By category, Home (5.13%) and Beauty (5.08%) have the highest return rates and Groceries (4.58%) the lowest.

Source in workbook: Pivots: Returns by Product; Return Rate by Category

7. Which supplier supplies the highest number of products?

> **Answer: No single supplier leads. Eleven suppliers tie at 2 products each, and the other 478 suppliers supply 1 product each.**

Evidence: The eleven are: Dyer LLC, Garcia PLC, Jones Group, Martinez Ltd, Matthews and Sons, Myers and Sons, Rosales PLC, Smith LLC, Smith PLC, Thomas Ltd and Williams Group (each listed as '... Supplies'). There are 489 suppliers for 500 products, so supply is spread very thinly, close to one supplier per product. The data has no delivery or quality fields, so supplier performance cannot be judged.

Source in workbook: Pivots: Products per Supplier

8. Which stores have low inventory?

> **Answer: STR006 (Kaduna) and STR014 (Port Harcourt) have the most low-stock products, 3 each. STR001 (Abuja), STR018 (Ibadan) and STR009 (Ibadan) have 2 each.**

Evidence: A product is counted as low when stock on hand is at or below its reorder level. Overall, 21 of 500 products (4.2%) are low, spread across 14 of the 25 stores; 11 stores have none. The biggest gaps below reorder level are Beauty Item 479 (10 in stock, level 29), Home Item 414 (13 vs 44), Home Item 212 (12 vs 41), Home Item 302 (14 vs 46) and Electronics Item 165 (14 vs 45). Each product is held at only one store, so stock transfers between stores cannot be supported by this data.

Source in workbook: Pivots: Inventory by Store

9. Which region performs best?

> **Answer: By total revenue, South performs best with ₦2,424,534,663 (32.0% of revenue), followed by East, North and West. Per store, the regions are essentially equal.**

Evidence: East ₦2.12B (28.0%), North ₦1.83B (24.2%), West ₦1.20B (15.9%). Revenue per store: North ₦305.1M, South ₦303.1M, East ₦302.7M, West ₦300.3M. South leads because it has 8 stores against 4 in West, not because each store sells more. Caution: The Region labels in the Stores table do not match the cities (the same city appears under different regions), so regional results should be confirmed with the client.

Source in workbook: Pivots: Sales by Region; Dashboard: Region slicer

10. Which product category should management invest in?

> **Answer: Electronics is the recommended category for investment, with Beauty as a second choice.**

Evidence: Electronics grew fastest (about +6% in the second half of the year against the first), has the highest average order value (₦153,343), a lower return rate (4.80%) than Beauty (5.08%) or Home (5.13%), and earns ₦15.4M per product against ₦14.8M for Beauty. Beauty is the largest category (23.3%) and grew about 4%. Home is the weakest: it is the only category that shrank (about -1%) and has the highest return rate. Caution: the gaps between categories are modest, growth rests on one year of data, and catalogue margins are similar (about 29-30%) for every category.

Source in workbook: Pivots: Sales by Category, Return Rate by Category; Dashboard

Definitions and assumptions used in these answers

- Revenue is the sum of Line_Total from Order_Items for all orders (Completed, Pending and Cancelled), because the brief gives no rule for order status. Total revenue is ₦7,575,250,086.
- Amounts are assumed to be Nigerian naira; the data contains no currency symbol.
- Discount is used exactly as supplied in Line_Total (Line_Total = Quantity x Unit_Price - Discount). The Discount values (0 to 20) look like percentages but are subtracted as flat amounts; this should be confirmed with the client.
- Return rate = unique returned orders / total orders (2,442 / 50,000 = 4.88%). 58 Order_IDs appear twice in Returns, and 1,252 returns are dated before their order date; both are source-data issues noted in the Data_Quality_Log.
- State and Region come from the store where the order was placed. Membership Level is the Customer_Segment field (Bronze, Silver, Gold).
- Stores are compared by Store_ID because the 25 stores share only 10 names.